Solar Fe I appendix

Sun · Fe I · downloadable appendix

Fe.json v1.390 · 125 Fe I products · 155 total

Solar Fe I anchor: 7.477 ± 0.052

HARPS · Reference Grade · Amarsi 3D-NLTE · n = 21. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.532 ± 0.079

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · n = 53

Uncertainty budget closed for six of eight rows (RYA-1226); two Fe I Kitt Peak/molecfit rows keep a legacy bar pending their microturbulence bracket. On the migrated rows most of the quoted systematic is a conservative profile-width term carried from a measurement in another band.

VIS

7.504 ± 0.049

Kitt Peak Molecfit · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

VIS

7.477 ± 0.052

HARPS · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

red-optical

7.421 ± 0.051

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · n = 66

NIR

7.482 ± 0.062

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · n = 20

H

7.497 ± 0.152

CRIRES+ · Synth · 1D-LTE · Codex Grade · n = 19

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Solid bars show statistical uncertainty from the line measurements; wireframes show systematic uncertainty from the product’s adopted error budget, which does not average away with more lines. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool’s provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

Reference Grade. An externally published reference line pool, using that reference’s line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a

promise that every line has an individual primary-laboratory uncertainty.

Codex Grade. The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.

Deep Grade. The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

3D-NLTE (Amarsi) includes three-dimensional atmospheric structure and non-LTE line formation; these products apply the published Amarsi correction grid to a 1D baseline.

Mean-3D / Frankenstein combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

Synthesis fits the spectral line profile. **EW** infers abundance from a line's integrated absorption (equivalent width). See Methodology for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

blue solid = statistical σ · wireframe = systematic σ · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders · N/A = no product for this model

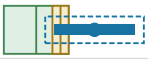
near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.29–7.82

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=55		7.554 ±0.061 stat ±0.237 syst
Synth · 1D-LTE · Gerber no product		N/A
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — Kurucz 2005 corrected Reference pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.29–7.82

EW · 1D-LTE no product		N/A

near-UV — Kitt Peak — Kurucz 2005 corrected Reference pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.29–7.82

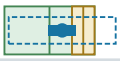
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber Reference Grade · n=53		7.532 ±0.050 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — 1984 Molecfit corrected Reference pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.29–7.82

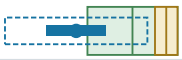
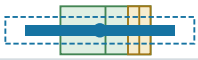
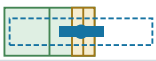
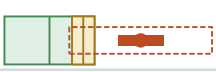
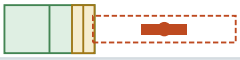
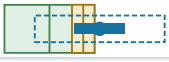
EW · 1D-LTE no product		N/A
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber Reference Grade · n=51		7.435 ±0.057 stat ±0.059 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV axis: 7.29 · 7.42 · 7.55 · 7.69 · 7.82 A(Fe) dex

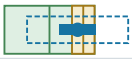
VIS — Kitt Peak — Kurucz 2005 corrected Asplund pool solar_kpno_kurucz2005_corrected A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber no product		N/A
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.471 ±0.012 stat ±0.048 syst

VIS — Kitt Peak — Kurucz 2005 corrected Codex graded pool solar_kpno_kurucz2005_corrected A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE Codex Grade · n=13 · 1 alternate		7.445 ±0.062 stat ±0.083 syst
Synth · 1D-LTE Codex Grade · n=65		7.410 ±0.027 stat ±0.063 syst
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.439 ±0.021 stat ±0.063 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.455 ±0.067 stat ±0.084 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=59		7.411 ±0.028 stat ±0.068 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=67		7.488 ±0.020 stat ±0.063 syst
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.541 ±0.020 stat ±0.063 syst
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.562 ±0.020 stat ±0.063 syst
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.505 ±0.022 stat ±0.058 syst

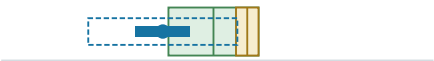
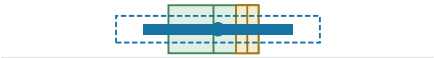
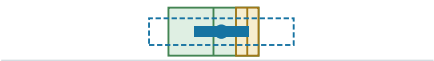
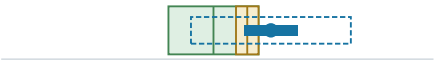
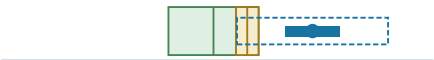
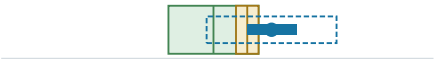
VIS — Kitt Peak — Kurucz 2005 corrected Reference pool solar_kpno_kurucz2005_corrected A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=166		7.367 ±0.015 stat ±0.065 syst
Synth · 1D-LTE · Gerber Reference Grade · n=167		7.373 ±0.015 stat ±0.067 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=156		7.371 ±0.016 stat ±0.070 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.430 ±0.015 stat ±0.061 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=101		7.485 ±0.016 stat ±0.045 syst

VIS — Kitt Peak — 1984 Molecfit corrected Asplund pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber no product		N/A
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.504 ±0.013 stat ±0.048 syst

VIS — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE Codex Grade · n=13		7.463 ±0.062 stat ±0.092 syst
Synth · 1D-LTE Codex Grade · n=65		7.414 ±0.025 stat ±0.062 syst
Synth · 1D-LTE · Gerber Codex Grade · n=65		7.415 ±0.024 stat ±0.066 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.464 ±0.066 stat ±0.091 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=59		7.420 ±0.026 stat ±0.068 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=65		7.467 ±0.024 stat ±0.064 syst
Synth · <3D>-LTE · Gerber · stagger Codex Grade · n=65		7.511 ±0.024 stat ±0.071 syst
Synth · <3D>-NLTE · Gerber · stagger Codex Grade · n=65		7.548 ±0.024 stat ±0.067 syst
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.022 stat ±0.058 syst

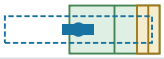
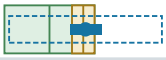
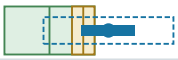
VIS — Kitt Peak — 1984 Molecfit corrected Reference pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=167		7.368 ±0.015 stat ±0.063 syst
Synth · 1D-LTE · Gerber Reference Grade · n=166		7.378 ±0.015 stat ±0.064 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=157		7.377 ±0.015 stat ±0.070 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.432 ±0.015 stat ±0.064 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=101		7.492 ±0.016 stat ±0.045 syst

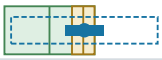
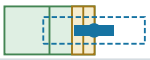
VIS — HARPS Asplund pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber no product		N/A
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.477 ±0.020 stat ±0.048 syst

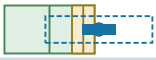
VIS — HARPS Codex graded pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE Codex Grade · n=6 · 1 alternate		7.498 ±0.070 stat ±0.097 syst
Synth · 1D-LTE Codex Grade · n=171		7.428 ±0.014 stat ±0.065 syst
Synth · 1D-LTE · Gerber Codex Grade · n=171		7.436 ±0.014 stat ±0.064 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=5		7.433 ±0.077 stat ±0.095 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=144		7.429 ±0.016 stat ±0.071 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=171		7.492 ±0.014 stat ±0.068 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.024 stat ±0.058 syst

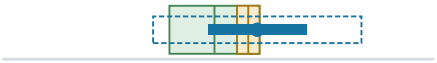
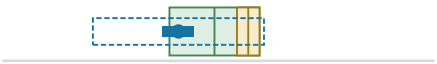
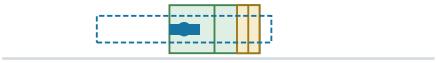
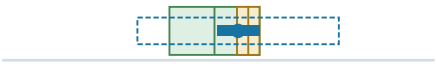
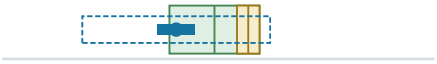
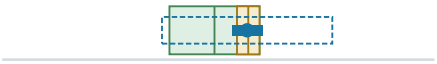
VIS — HARPS Reference pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=167		7.428 ±0.018 stat ±0.066 syst
Synth · 1D-LTE · Gerber Reference Grade · n=167		7.441 ±0.018 stat ±0.066 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=157		7.439 ±0.018 stat ±0.071 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.491 ±0.017 stat ±0.065 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=100		7.500 ±0.017 stat ±0.045 syst

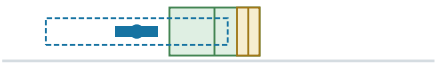
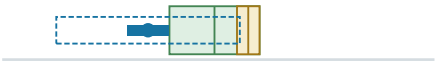
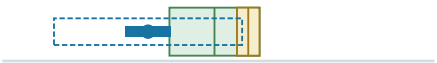
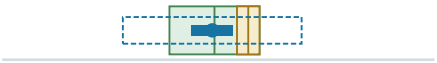
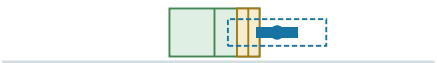
VIS — IAG Asplund pool solar_iag
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber no product		N/A
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.504 ±0.015 stat ±0.048 syst

VIS — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE Codex Grade · n=5		7.498 ±0.044 stat ±0.093 syst
Synth · 1D-LTE Codex Grade · n=131		7.428 ±0.014 stat ±0.076 syst
Synth · 1D-LTE · Gerber Codex Grade · n=131		7.433 ±0.013 stat ±0.078 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=4		7.481 ±0.019 stat ±0.089 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=103		7.426 ±0.017 stat ±0.083 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=131		7.489 ±0.014 stat ±0.076 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=39		7.543 ±0.025 stat ±0.056 syst

VIS — IAG Reference pool solar_iag
A(Fe) dex; axis 7.27–7.66

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=81		7.391 ±0.019 stat ±0.081 syst
Synth · 1D-LTE · Gerber Reference Grade · n=81		7.401 ±0.018 stat ±0.081 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=71		7.401 ±0.020 stat ±0.084 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=81		7.458 ±0.018 stat ±0.079 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=78		7.516 ±0.019 stat ±0.044 syst

VIS axis: 7.27 · 7.37 · 7.46 · 7.56 · 7.66 A(Fe) dex

red-optical — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.26–7.65

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=97		7.437 ±0.013 stat ±0.052 syst
Synth · 1D-LTE · Gerber Codex Grade · n=97		7.440 ±0.013 stat ±0.053 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=46		7.449 ±0.017 stat ±0.060 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=97		7.487 ±0.014 stat ±0.055 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

red-optical — Kitt Peak — Kurucz 2005 corrected Reference pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.26–7.65

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=68		7.423 ±0.017 stat ±0.052 syst
Synth · 1D-LTE · Gerber Reference Grade · n=66		7.421 ±0.017 stat ±0.048 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=34		7.418 ±0.018 stat ±0.063 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=68		7.455 ±0.017 stat ±0.052 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

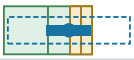
red-optical — Kitt Peak — 1984 Molecfit corrected Codex graded pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.26–7.65

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=66		7.413 ±0.017 stat ±0.053 syst
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.419 ±0.017 stat ±0.054 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=33		7.403 ±0.020 stat ±0.067 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=65		7.454 ±0.017 stat ±0.166 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

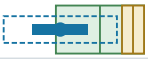
red-optical — Kitt Peak — 1984 Molecfit corrected Reference pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.26–7.65

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=66		7.413 ±0.017 stat ±0.053 syst
Synth · 1D-LTE · Gerber Reference Grade · n=67		7.419 ±0.017 stat ±0.054 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=33		7.403 ±0.020 stat ±0.067 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=65		7.454 ±0.017 stat ±0.166 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

red-optical — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.26–7.65

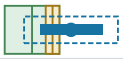
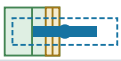
EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=96		7.437 ±0.020 stat ±0.056 syst
Synth · 1D-LTE · Gerber Codex Grade · n=96		7.437 ±0.020 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=46		7.446 ±0.034 stat ±0.064 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=95		7.479 ±0.020 stat ±0.055 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

red-optical — IAG Reference pool solar_iag
A(Fe) dex; axis 7.26–7.65

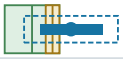
EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=67		7.424 ±0.025 stat ±0.051 syst
Synth · 1D-LTE · Gerber Reference Grade · n=66		7.423 ±0.025 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=33		7.422 ±0.043 stat ±0.061 syst
Synth · 1D-NLTE · Gerber Reference Grade · n=66		7.454 ±0.025 stat ±0.051 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

red-optical axis: 7.26 · 7.36 · 7.45 · 7.55 · 7.65 A(Fe) dex

NIR — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.23–7.86

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=21		7.517 ±0.045 stat ±0.069 syst
Synth · 1D-LTE · Gerber Codex Grade · n=21		7.508 ±0.046 stat ±0.077 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=5		7.538 ±0.088 stat ±0.131 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — Kitt Peak — Kurucz 2005 corrected Reference pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.23–7.86

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=21		7.517 ±0.045 stat ±0.069 syst
Synth · 1D-LTE · Gerber Reference Grade · n=21		7.508 ±0.046 stat ±0.077 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.538 ±0.088 stat ±0.131 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

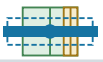
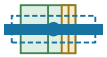
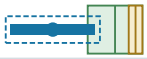
NIR — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.23–7.86

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=18		7.495 ±0.036 stat ±0.058 syst
Synth · 1D-LTE · Gerber Codex Grade · n=20		7.482 ±0.033 stat ±0.052 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=5		7.368 ±0.061 stat ±0.112 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

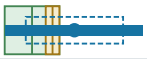
NIR — Kitt Peak — 1984 Molecfit corrected Reference pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.23–7.86

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=18		7.495 ±0.036 stat ±0.058 syst
Synth · 1D-LTE · Gerber Reference Grade · n=20		7.482 ±0.033 stat ±0.052 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.368 ±0.061 stat ±0.112 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.23–7.86

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=24		7.460 ±0.069 stat ±0.062 syst
Synth · 1D-LTE · Gerber Codex Grade · n=23		7.468 ±0.072 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=7		7.369 ±0.062 stat ±0.069 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — IAG Reference pool solar_iag
A(Fe) dex; axis 7.23–7.86

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=17		7.520 ±0.095 stat ±0.070 syst
Synth · 1D-LTE · Gerber Reference Grade · n=16		7.522 ±0.101 stat ±0.071 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.515 ±0.106 stat ±0.080 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — CRIRES+ Codex graded pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.23–7.86

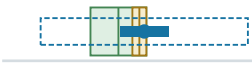
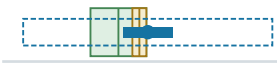
EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=5		7.551 ±0.088 stat ±0.063 syst
Synth · 1D-LTE · Gerber Codex Grade · n=5		7.552 ±0.087 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=2		7.637 ±0.192 stat ±0.164 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — CRIRES+ Reference pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.23–7.86

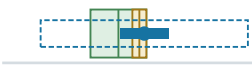
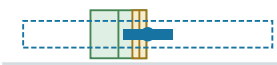
EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=5		7.551 ±0.088 stat ±0.063 syst
Synth · 1D-LTE · Gerber Reference Grade · n=5		7.552 ±0.087 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.637 ±0.192 stat ±0.164 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR axis: 7.23 · 7.38 · 7.54 · 7.70 · 7.86 A(Fe) dex

H — CRIRES+ Codex graded pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.29–7.91

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=19		7.497 ±0.035 stat ±0.148 syst
Synth · 1D-LTE · Gerber Codex Grade · n=19		7.502 ±0.035 stat ±0.178 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber Codex Grade · n=8		7.649 ±0.029 stat ±0.233 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

H — CRIRES+ Reference pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.29–7.91

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=19		7.497 ±0.035 stat ±0.148 syst
Synth · 1D-LTE · Gerber Reference Grade · n=19		7.502 ±0.035 stat ±0.178 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber Reference Grade · n=8		7.649 ±0.029 stat ±0.233 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

H axis: 7.29 · 7.45 · 7.60 · 7.76 · 7.91 A(Fe) dex

Read all product values, including alternate line sets and experiments

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · 1D-LTE: 7.497 ± 0.152 reported total; n=19; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · ENGINE-B-NLTE: 7.649 ± 0.235 reported total; n=8; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.502 ± 0.182 reported total; n=19; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · 1D-LTE: 7.497 ± 0.152 reported total; n=19; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.649 ± 0.235 reported total; n=8; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.502 ± 0.182 reported total; n=19; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · 1D-LTE: 7.551 ± 0.108 reported total; n=5; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · ENGINE-A: 7.637 ± 0.252 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.552 ± 0.107 reported total; n=5; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · 1D-LTE: 7.551 ± 0.108 reported total; n=5; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · ENGINE-A: 7.637 ± 0.252 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.552 ± 0.107 reported total; n=5; ξ MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.460 ± 0.093 reported total; n=24; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.369 ± 0.093 reported total; n=7; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.468 ± 0.094 reported total; n=23; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.520 ± 0.118 reported total; n=17; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.515 ± 0.133 reported total; n=4; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.522 ± 0.123 reported total; n=16; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.517 ± 0.082 reported total; n=21; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.538 ± 0.158 reported total; n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.508 ± 0.090 reported total; n=21; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.517 ± 0.082 reported total; n=21; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.538 ± 0.158 reported total; n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.508 ± 0.090 reported total; n=21; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.495 ± 0.068 reported total; n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.368 ± 0.127 reported total; n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.482 ± 0.062 reported total; n=20; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.495 ± 0.068 reported total; n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.368 ± 0.127 reported total; n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.482 ± 0.062 reported total; n=20; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfit_corrected · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.477 ± 0.052 reported total; n=21; ξ NOT_APPLICABLE

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED-LOCALRENORM · SYNTH · 1D-LTE: 7.339 ± 0.087 reported total; n=109; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED-LOCALRENORM · SYNTH · ENGINE-A: 7.350 ± 0.088 reported total; n=105; ξ MEASURED

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.419 ± 0.077 reported total; n=106; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.429 ± 0.083 reported total; n=102; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.498 ± 0.120 reported total; n=6; ξ MEASURED

HARPS · EW · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.433 ± 0.123 reported total; n=5; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.431 ± 0.122 reported total; n=5; ξ MEASURED

HARPS · Synth · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.428 ± 0.067 reported total; n=171; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.062 reported total; n=50; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.429 ± 0.072 reported total; n=144; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.492 ± 0.069 reported total; n=171; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.436 ± 0.066 reported total; n=171; ξ MEASURED

HARPS · Synth · 1D-LTE · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.428 ± 0.068 reported total; n=167; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.500 ± 0.048 reported total; n=100; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.439 ± 0.073 reported total; n=157; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.491 ± 0.067 reported total; n=167; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.441 ± 0.069 reported total; n=167; ξ MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_iag · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.504 ± 0.050 reported total; n=21; ξ NOT_APPLICABLE

IAG · Synth · 1D-LTE · Deep Grade · solar_iag · DEEPGRADED · SYNTH · 1D-LTE: 7.384 ± 0.082 reported total; n=41; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_iag · DEEPGRADED · SYNTH · ENGINE-A: 7.391 ± 0.083 reported total; n=37; ξ MEASURED

IAG · EW · 1D-LTE · Codex Grade · solar_iag · GRADED · PROFILEFIT · 1D-LTE: 7.498 ± 0.102 reported total; n=5; ξ MEASURED

IAG · EW · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · PROFILEFIT · ENGINE-A: 7.481 ± 0.091 reported total; n=4; ξ MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.428 ± 0.077 reported total; n=131; ξ MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.543 ± 0.061 reported total; n=39; ξ NOT_APPLICABLE

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.426 ± 0.085 reported total; n=103; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-B-NLTE: 7.489 ± 0.077 reported total; n=131; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.433 ± 0.079 reported total; n=131; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.391 ± 0.083 reported total; n=81; ξ MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.516 ± 0.047 reported total; n=78; ξ NOT_APPLICABLE

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.401 ± 0.086 reported total; n=71; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.458 ± 0.081 reported total; n=81; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.401 ± 0.084 reported total; n=81; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.471 ± 0.049 reported total; n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.329 ± 0.068 reported total; n=106; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.340 ± 0.075 reported total; n=101; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.445 ± 0.104 reported total; n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.455 ± 0.107 reported total; n=12; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.479 ± 0.090 reported total; n=9; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.410 ± 0.069 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.505 ± 0.062 reported total; n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.411 ± 0.074 reported total; n=59; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.488 ± 0.066 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.439 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.541 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.066 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.367 ± 0.067 reported total; n=166; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.485 ± 0.048 reported total; n=101; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.371 ± 0.071 reported total; n=156; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.430 ± 0.063 reported total; n=167; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.373 ± 0.068 reported total; n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfit_corrected · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.504 ± 0.049 reported total; n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.347 ± 0.073 reported total; n=106; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.357 ± 0.078 reported total; n=102; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.463 ± 0.111 reported total; n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.464 ± 0.112 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.414 ± 0.067 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.062 reported total; n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.420 ± 0.073 reported total; n=59; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.467 ± 0.069 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.415 ± 0.070 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.511 ± 0.075 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.548 ± 0.071 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.368 ± 0.065 reported total; n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.492 ± 0.048 reported total; n=101; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.377 ± 0.072 reported total; n=157; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.432 ± 0.066 reported total; n=167; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.378 ± 0.066 reported total; n=166; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.554 ± 0.245 reported total; n=55; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.532 ± 0.079 reported total; n=53; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.435 ± 0.082 reported total; n=51; ξ MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.437 ± 0.059 reported total; n=96; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.446 ± 0.073 reported total; n=46; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-B-NLTE: 7.479 ± 0.059 reported total; n=95; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.437 ± 0.064 reported total; n=96; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.424 ± 0.057 reported total; n=67; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.422 ± 0.075 reported total; n=33; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.057 reported total; n=66; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.423 ± 0.066 reported total; n=66; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.437 ± 0.054 reported total; n=97; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.449 ± 0.063 reported total; n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.487 ± 0.057 reported total; n=97; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.440 ± 0.055 reported total; n=97; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.423 ± 0.054 reported total; n=68; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.418 ± 0.066 reported total; n=34; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.455 ± 0.055 reported total; n=68; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.421 ± 0.051 reported total; n=66; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.413 ± 0.056 reported total; n=66; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.403 ± 0.070 reported total; n=33; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.167 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.419 ± 0.056 reported total; n=67; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.413 ± 0.056 reported total; n=66; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.403 ± 0.070 reported total; n=33; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.167 reported total; n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.419 ± 0.056 reported total; n=67; ξ MEASURED

Solar iron context

Fe I supplies the bulk of the neutral-iron line statistics used for the Solar abundance reference. Separate instruments, line pools and treatments test the stability of that reference; their differences remain visible.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline. The headline is the per-band best-available product, selected from this feed by the publication generator rather than fixed as a constant: RYA-850 promoted the graded laboratory-gf pool to the PRIMARY reported value, and RYA-819 retired the earlier constructed anchor after showing its Magic-2013 correction wrong in both magnitude and shape. RYA-161: nothing calibrated to a target.

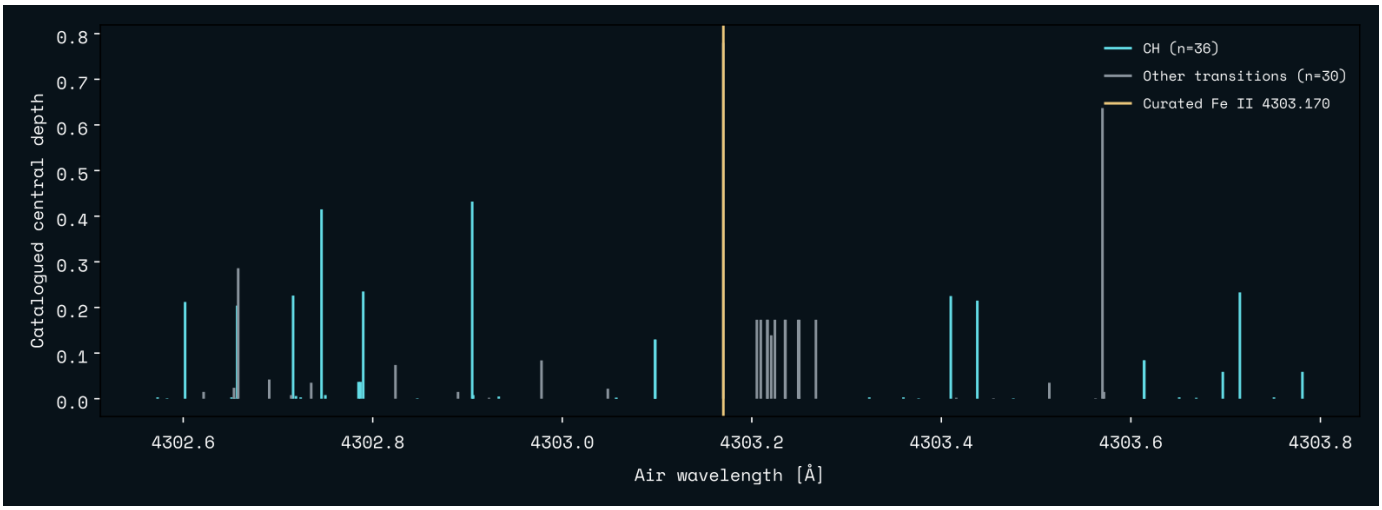
Frankenstein — experimental, pending the Bride

2 experimental products remain available in the Fe I product values and downloads.

RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

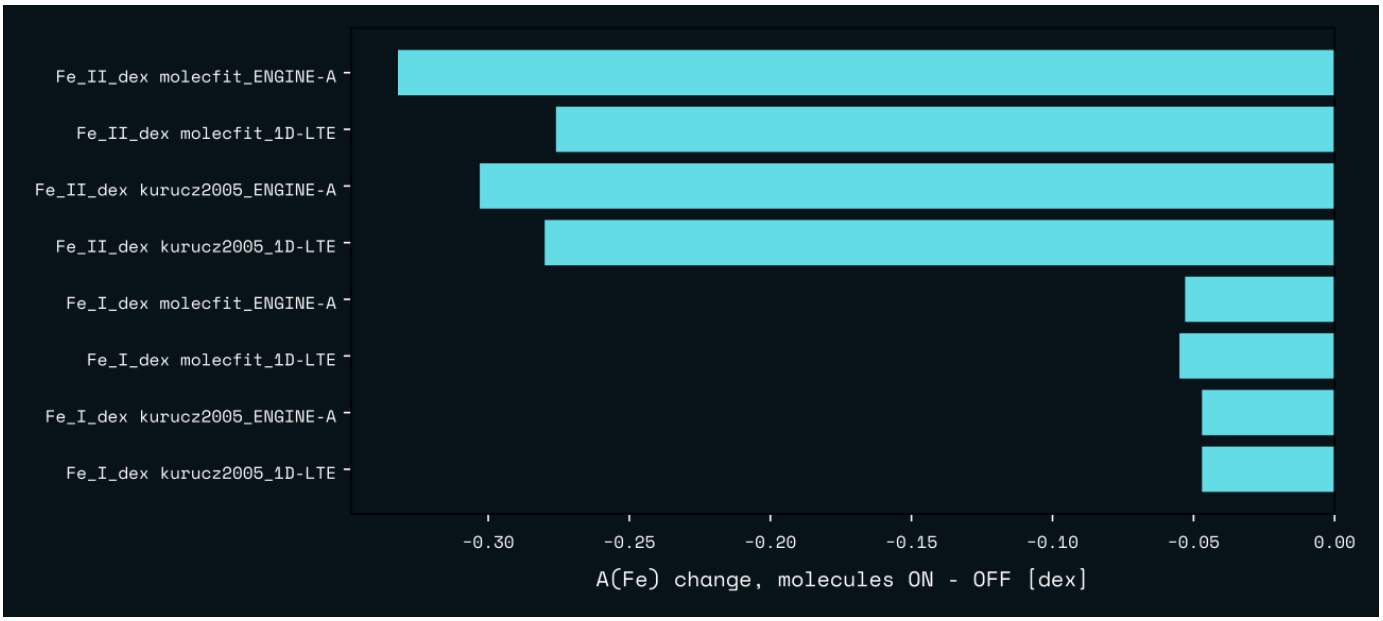
Problem-line and model diagnostics

CH G-band artifact



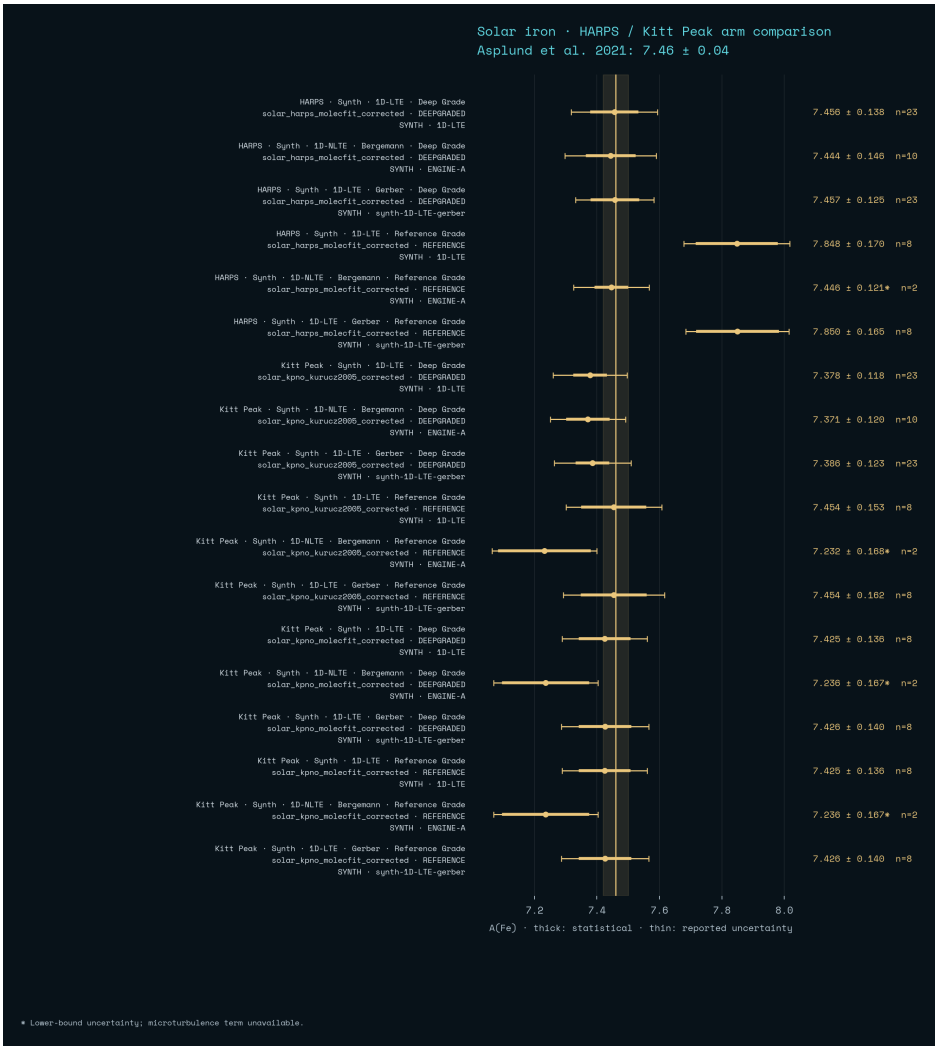
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · A = 10.0 (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 A fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator’s fit window; it is not a synthetic spectrum. Source evidence

Near-UV molecular opacity



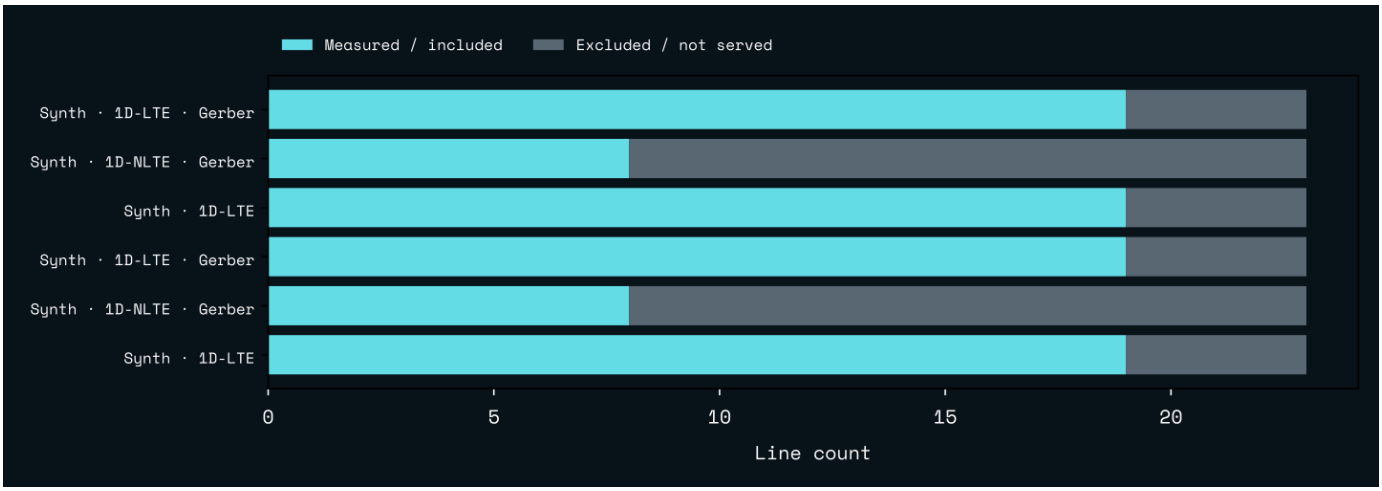
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. Source evidence

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. Source evidence

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. Source evidence

Download the evidence

809 graded per-line measurements exported. Evidence for 137 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

Graded per-line abundances (CSV)

All product fields (CSV; nested fields preserved as JSON)

Per-product line-evidence coverage (CSV)

Source Fe.json

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0
Science source commit c93d65e62c04a990f59dfb57bdb0b41f87966c58
Fe.json v1.390 · feed timestamp 2026-10-19T05:21:12Z
Generated 2026-10-19T05:21:38+00:00
Feed SHA-256 1308bb0ac396bbd99db4a12ee2572193bcf54318bda29aa291ea0f8303819591

Refreshed product tracker · Build manifest · Fe II appendix

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Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Amarsi, A. M.; Liljegren, S.; Nissen, P. E. (2022). 3D non-LTE iron abundances in FG-type dwarfs (doi: 10.1051/0004-6361/202244542). The Amarsi 3D non-LTE iron correction grid used by the displayed products.

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Den Hartog, E. A.; Ruffoni, M. P.; Lawler, J. E.; Pickering, J. C.; Lind, K.; Brewer, N. R. (2014). Fe I Oscillator Strengths for Transitions from High-lying Even-parity Levels (doi: 10.1088/0067-0049/215/2/23). Laboratory

transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Brewer, N. R.; Lind, K.; Nave, G.; Pickering, J. C. (2014). Fe I oscillator strengths for the Gaia-ESO survey (doi: 10.1093/mnras/stu780). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Belmonte, M. T.; Pickering, J. C.; Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Guzman, A.; Heiter, U. (2017). Fe I Oscillator Strengths for Transitions from High-lying Odd-parity Levels (doi: 10.3847/1538-4357/aa8cd3). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.